Filbin et al. 2018 — Pediatric H3-K27M Mutant Glioma (Brain)

SmartSeq2 (TPM, log1p) · 2,118 malignant cells · 5 samples (BCH836/869/1126, MUV05/10) · run Filbin2018_Brain-6

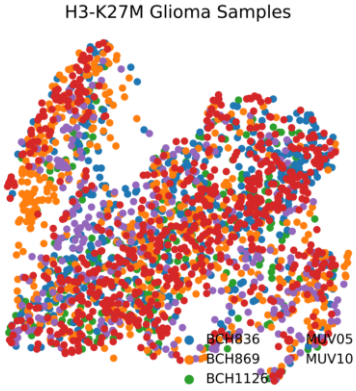
Question: do malignant cells recapitulate embryonic-development programs (gastrulation / posterior somitogenesis / neural crest)?

GO / NO-GO GATE LEDGER

Gate 1 — mixing + continuity	**GO**	Harmony θ=10 on `sample`; per-neighbourhood iLISI rises 1.0→3.66 (of 5); one connected kNN component. The OPC→AC/OC glial hierarchy is ONE continuous manifold, not per-sample islands.
Gate 2 — module geography	**GO**	Embryonic modules present & organized (pSom μ=+0.23). Gast & pSom CO-VARY (per-cell r=+0.27) and overlap across all arms; Crest a weak inverse gradient. Overlapping modules present → GO.
Gate 3 — hub cross-repression	**NO-GO**	Manifold branches GLIALLY (stable 3-tip/1-fork PPT tree, fold-free) so Gate 3 is reached, but the node-cluster Gast↔pSom r=+0.59 and hub per-gene r=+0.17 are POSITIVE. The one negative r (−0.48) is a joint-score COLLIDER artifact — independence null already −0.85. No repression → NO-GO.

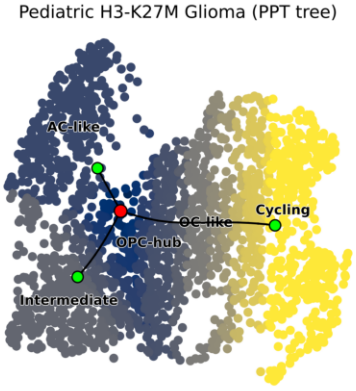
Geometry: BRANCHING (glial OPC→AC/OC fork) — but NO embryonic-module bifurcation. Root = OPC-hub (Gast/pSom-high, Crest-low origin).

Samples (Gate 1)



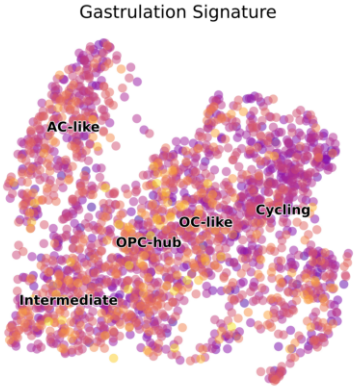
Filbin et al., 2018

Manifold + PPT tree (root=OPC-hub)



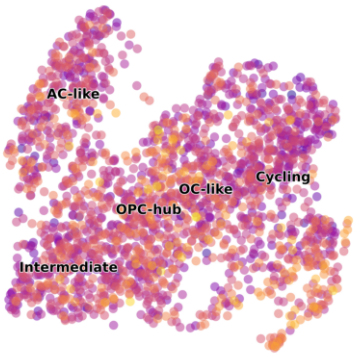
Filbin et al., 2018

Gastrulation



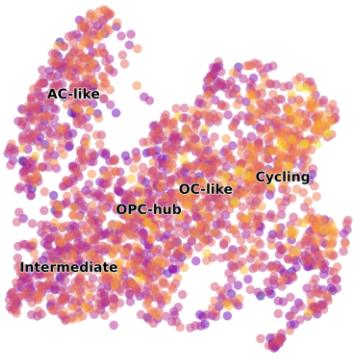
Filbin et al., 2018

Somitogenesis



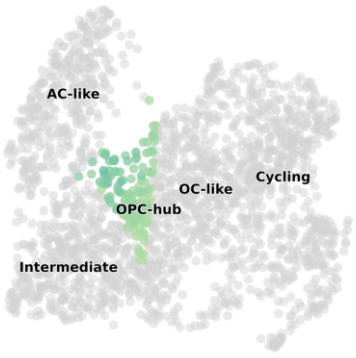
Filbin et al., 2018

Neural Crest



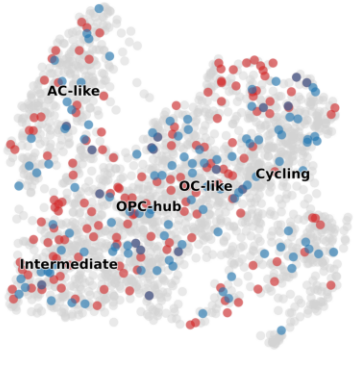
Filbin et al., 2018

Module corr. (Gate 3)



Filbin et al., 2018

EMT-TF expression



Filbin et al., 2018

PARAMETER CHOICES & RATIONALE

Sample threshold: ≥184 → 5/6, 2,118 cells
largest count gap 241→141 (ratio 1.71)

n_pcs / θ: 20 / 10
elbow rejected (under-powers); θ by iLISI sweep, no collapse

Trajectory: PPT tree, Nodes=30 λ=1000
stable 3-tip/1-fork (λ 500-1000); fold-free ratio 1.16

Root: OPC-hub fork (Gast/pSom-high)
developmental order: Gast+pSom max, Crest-low; NOT cycling/Crest tip

Hub: OPC-like fork (node 24, n=133)
topological convergence at the branch point

Hub r (Gast↔pSom): +0.17 hub / +0.59 nodes
POSITIVE; −0.48 joint-score = collider (null −0.85)

EMT-TFs shown: PRRX1, ZEB2 (VIM excluded)
ZEB1/ZEB2/PRRX1 detectable, all pt-flat; VIM is a marker

VERDICT: The embryonic pattern does NOT reproduce as a bifurcation. A genuine GLIAL bifurcation (OPC→AC/OC) exists, but Gast/pSom co-vary across it rather than cross-repress — the negative "hub r" is a joint-score collider artifact. Non-branching embryonic biology on a glially-branching manifold.